Fatima Jinnah Women University

Opening Portals of Excellence Through Higher Education

Cloud Computing

BSE-VB

<u>Submitted By</u>

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<u>Submitted to</u>

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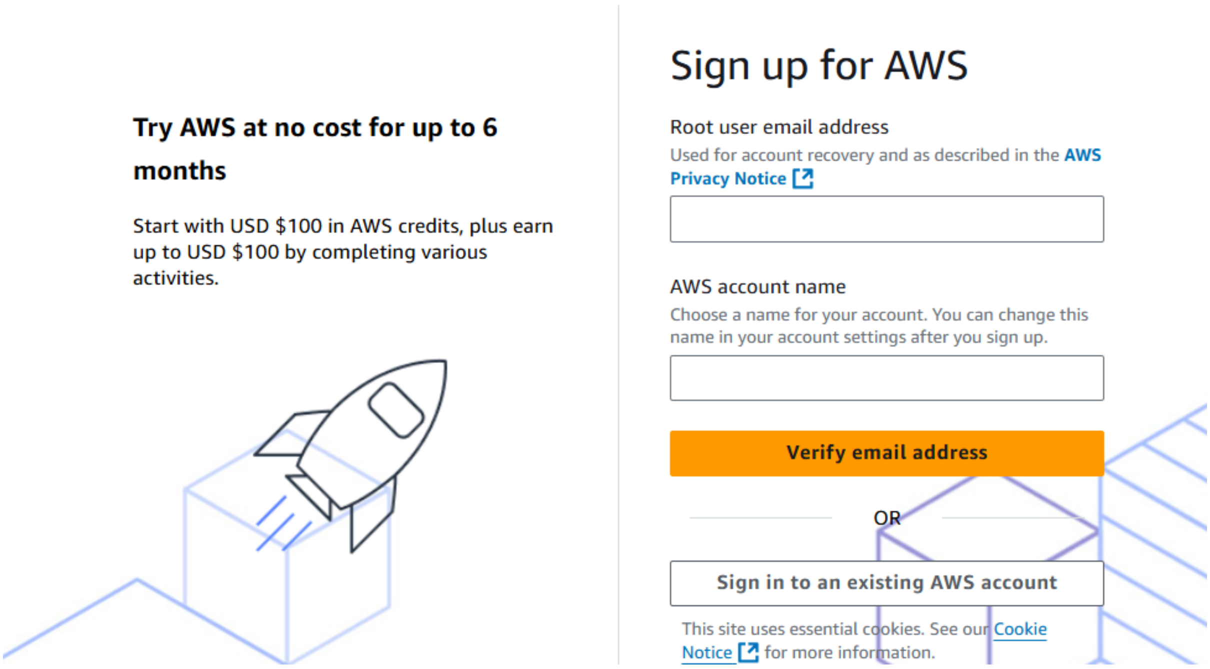
<u>LAB-08</u>

Task 1 — Create an AWS account and enable UAE (me-central-1)

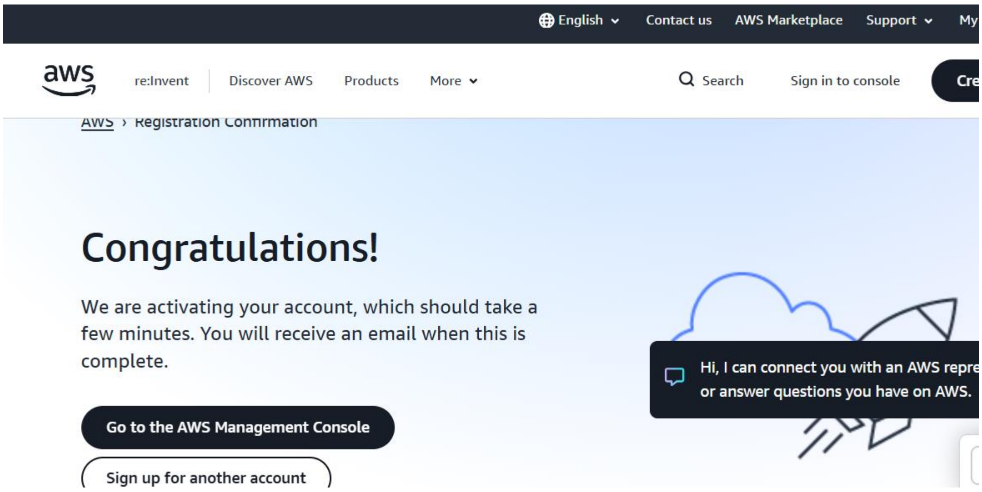
Step 1: Open AWS Signup Page

Screenshot: task1_open_signup_page.png



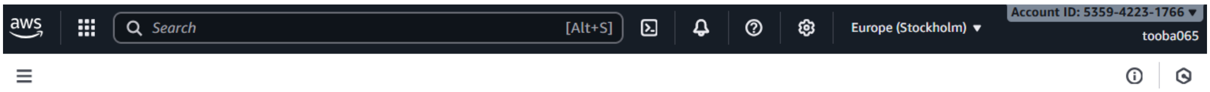
Step 2: Complete Registration

Screenshot: task1_signed_up_confirmation.png



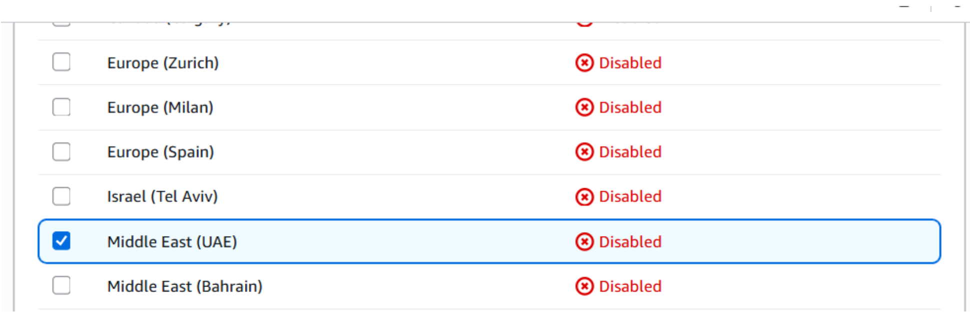
Step 3: Sign in as Root User

Screenshot: task1_root_signed_in.png

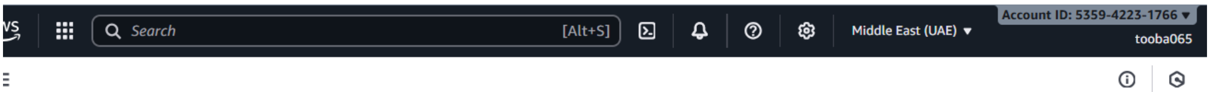


Step 4: Enable UAE Region (me-central-1)

Screenshot: task1_enable_region_me-central-1.png



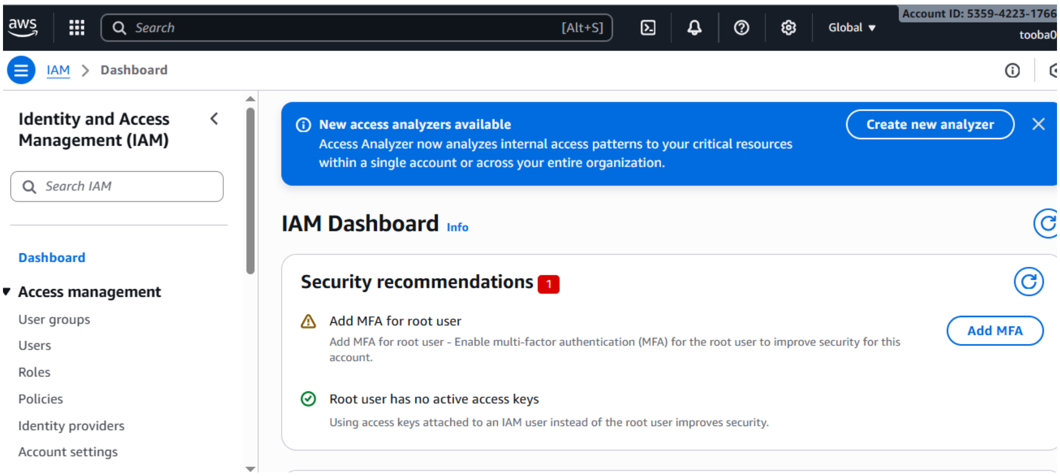
Step 5: Task 1 Summary — root console header + region active
Screenshot: task1_summary.png



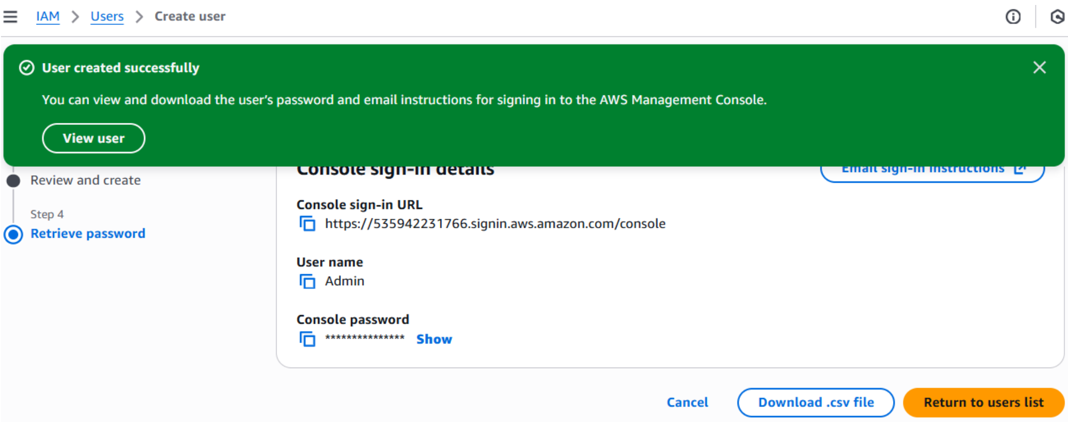
Task 2 — Create IAM Admin and Lab8User with Console Access

Goal: Create Admin IAM user, then Lab8User, both with AdministratorAccess.

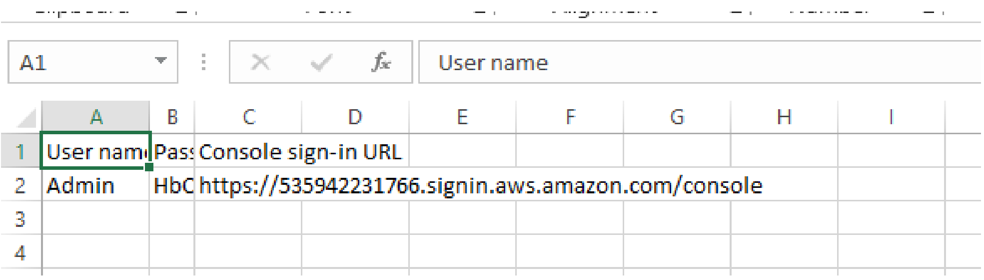
Step 1: Open IAM Console
Screenshot: task2_open_iam_console.png



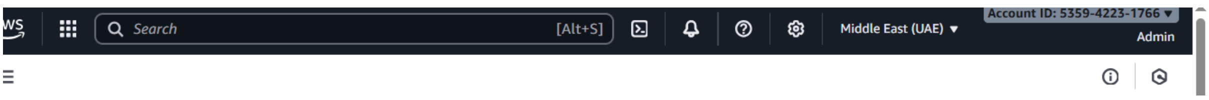
Step 2: Create Admin User (console access + AdministratorAccess)
Screenshot: task2_admin_create_confirmation.png



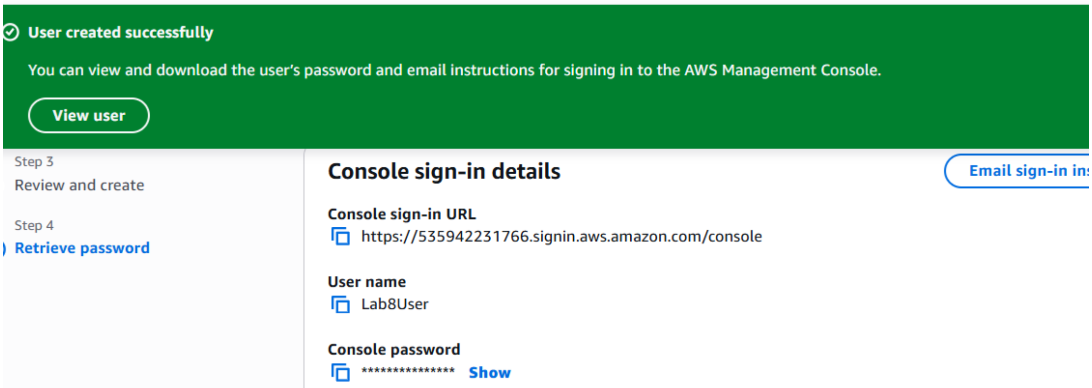
Step 3: Download Admin CSV and show on host
Screenshot: task2_admin_csv_and_signin_url.png



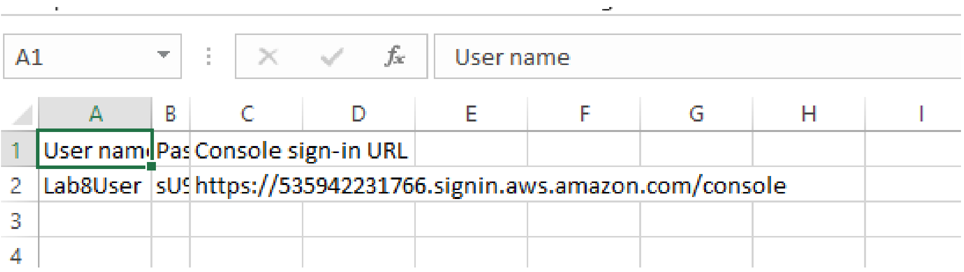
Step 4: Sign in as Admin
Screenshot: task2_admin_console_after_login.png



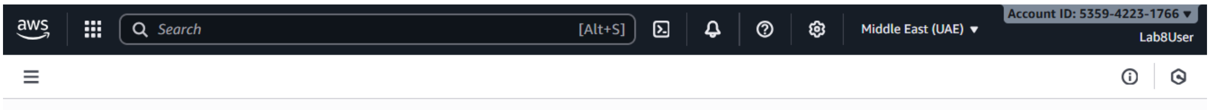
Step 5: Create Lab8User (console access + AdministratorAccess)
Screenshot: task2_create_lab8user_and_csv.png



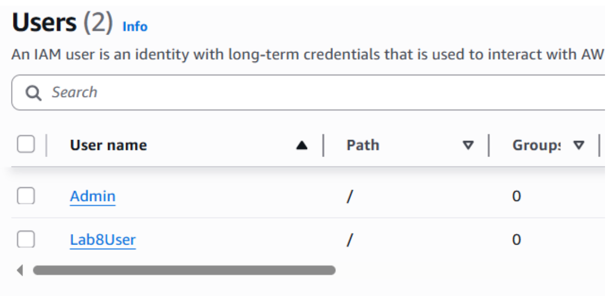
Step 6: Download Lab8User CSV on host
Screenshot: task2_lab8user_csv_saved.png



Step 7: Sign in as Lab8User
Screenshot: task2_lab8user_logged_in.png



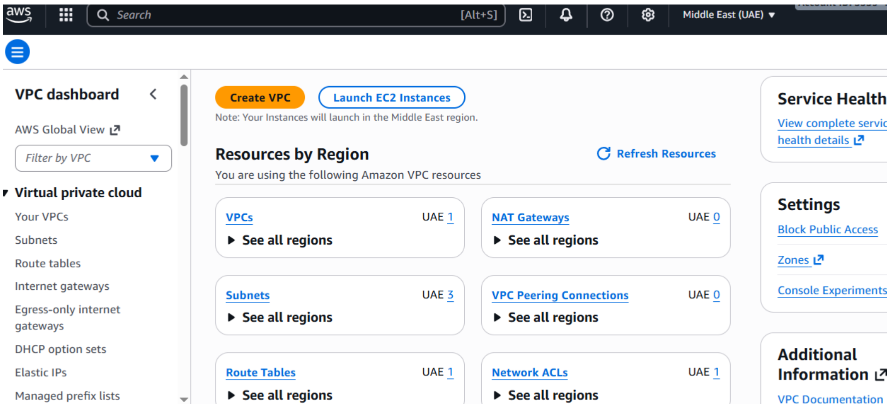
Step 8: Task 2 Summary — show both IAM users
Screenshot: task2_summary.png



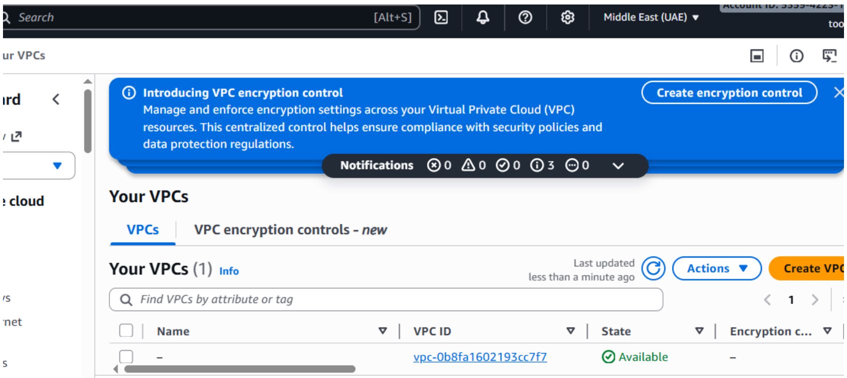
Task 3 — Inspect VPC Resources (me-central-1)

Goal: View counts of VPCs, Subnets, Route Tables, and Network ACLs in me-central-1.

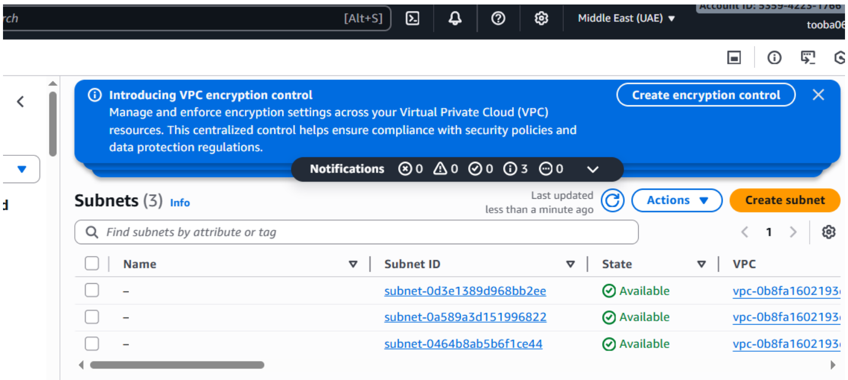
Step 1: Open VPC Console
Screenshot: task3_open_vpc_console.png



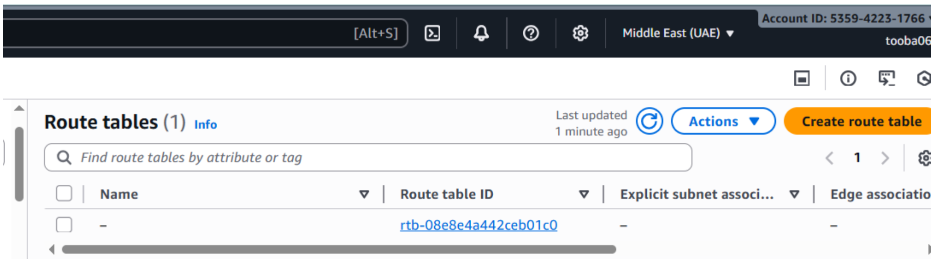
Step 2: View VPCs List
Screenshot: task3_vpcs_list.png



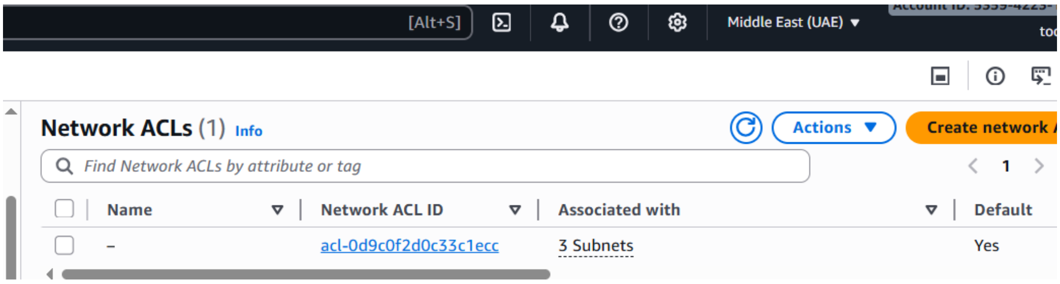
Step 3: View Subnets List
Screenshot: task3_subnets_list.png



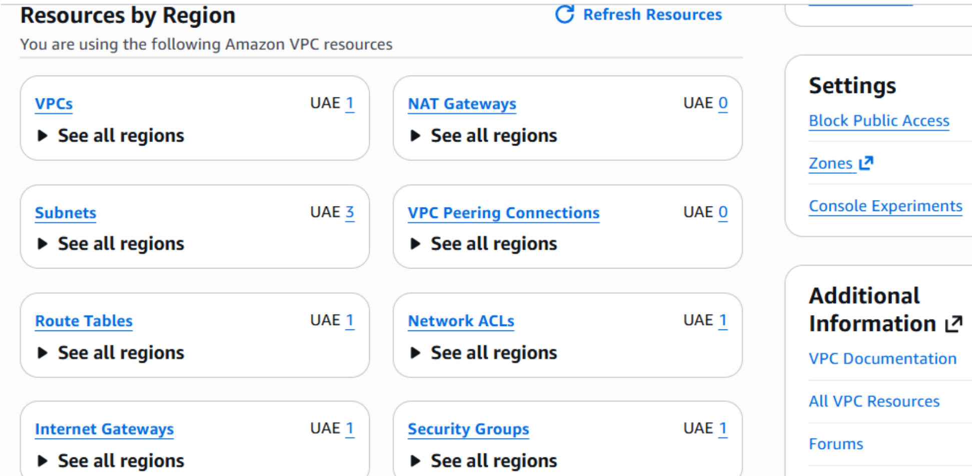
Step 4: View Route Tables List
Screenshot: task3_route_tables_list.png



Step 5: View Network ACLs List
Screenshot: task3_network_acls_list.png



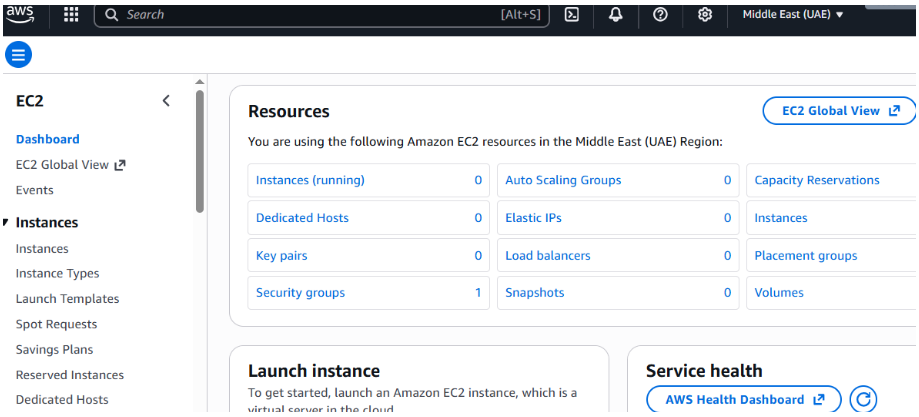
Step 6: Task 3 Summary — single screenshot showing all resources
Screenshot: task3_summary.png



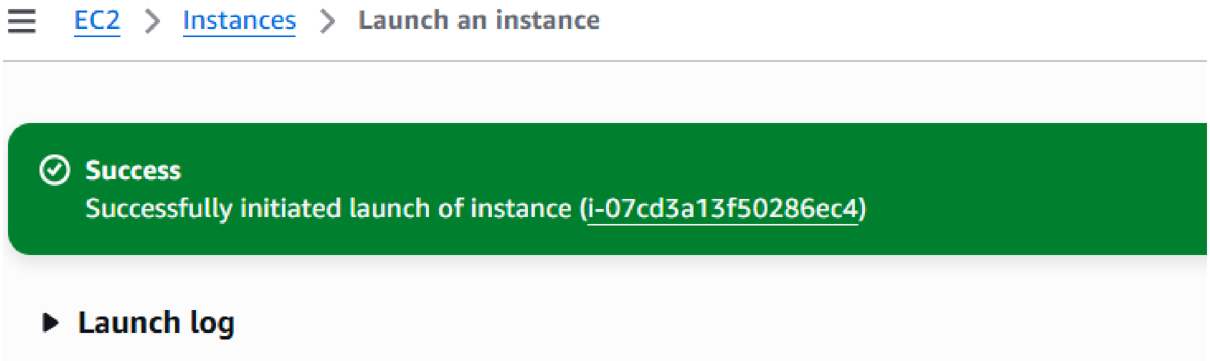
Task 4 — Launch EC2, SSH, Install Docker & Docker Compose, Deploy Gitea

Goal: Launch EC2 instance Lab8Machine, configure security, install Docker/Docker Compose, deploy Gitea, allow port 3000.

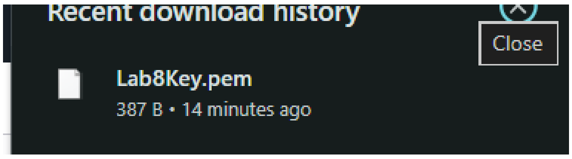
Step 1: Open EC2 Console (me-central-1)
Screenshot: task4_open_ec2_console.png



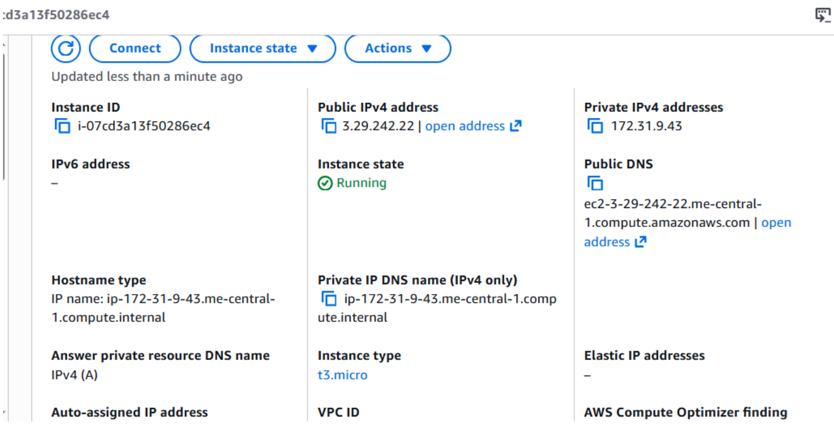
Step 2: Launch Instance Configuration
Screenshot: task4_launch_instance_config.png



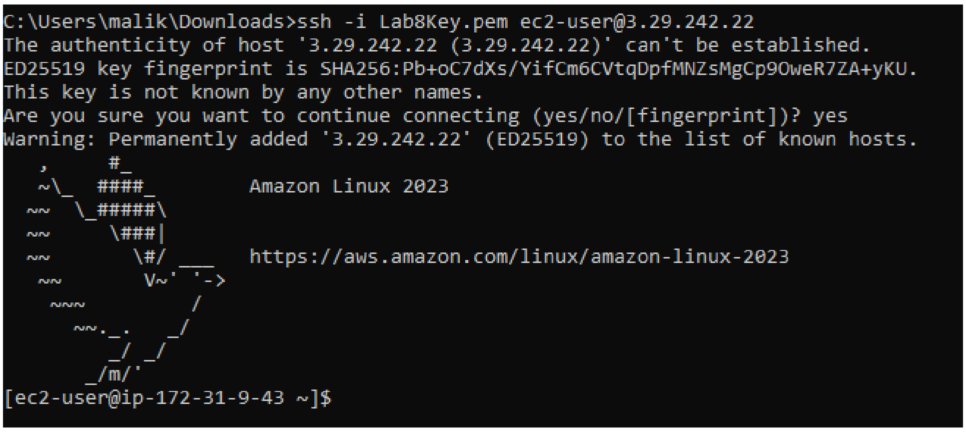
Step 3: Download Key Pair Lab8Key.pem
Screenshot: task4_keypair_download.png



Step 4: EC2 Instance running and public IPv4 visible
Screenshot: task4_instance_running_console.png



Step 5: SSH from Windows host to EC2 instance
Screenshot: task4_ssh_from_windows_to_ec2.png



Step 6: Install Docker & Docker Compose on EC2
Screenshot: task4_ec2_install_docker_compose_started.png

```
[ec2-user@ip-172-31-9-43 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-9-43 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository                    292 kB/s |  29 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-9-43 ~]$ sudo yum install -y docker
Last metadata expiration check: 0:00:55 ago on Thu Dec  4 16:14:24 2025.
Dependencies resolved.
================================================================================================
 Package                   Architecture   Version                      Repository          Size
================================================================================================
Installing:
 docker                    x86_64         25.0.13-1.amzn2023.0.2       amazonlinux         46 M
Installing dependencies:
 container-selinux         noarch         4:2.242.0-1.amzn2023         amazonlinux         58 k
 containerd                x86_64         2.1.4-1.amzn2023.0.2         amazonlinux         23 M
 iptables-libs             x86_64         1.8.8-3.amzn2023.0.2         amazonlinux        401 k
 iptables-nft              x86_64         1.8.8-3.amzn2023.0.2         amazonlinux        183 k
 libcgroup                 x86_64         3.0-1.amzn2023.0.1           amazonlinux         75 k
 libnetfilter_conntrack    x86_64         1.0.8-2.amzn2023.0.2         amazonlinux         58 k
 libnfnetlink              x86_64         1.0.1-19.amzn2023.0.2        amazonlinux         30 k
 libnftnl                  x86_64         1.2.2-2.amzn2023.0.2         amazonlinux         84 k
 pigz                      x86_64         2.5-1.amzn2023.0.3           amazonlinux         83 k
```

```
[ec2-user@ip-172-31-9-43 ~]$ sudo mkdir -p /usr/local/lib/docker/cli-plugins
[ec2-user@ip-172-31-9-43 ~]$ sudo curl -SL https://github.com/docker/compose/releases/latest/downloa
ux-x86_64 -o /usr/local/lib/docker/cli-plugins/docker-compose
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 29.8M  100 29.8M    0     0  33.7M      0 --:--:-- --:--:-- --:--:-- 33.7M
[ec2-user@ip-172-31-9-43 ~]$ sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose
[ec2-user@ip-172-31-9-43 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-9-43 ~]$
```

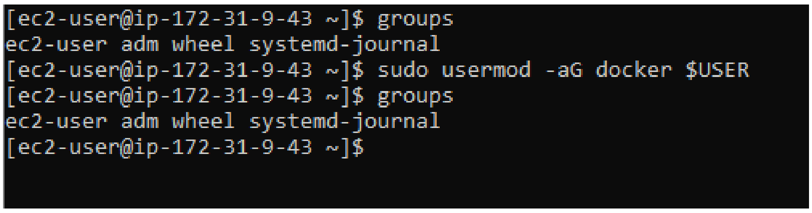
Step 7: Create/edit compose.yaml with Gitea content
Screenshot: task4_vim_compose_yaml_paste.png

```
      - POSTGRES_USER=gitea
      - POSTGRES_PASSWORD=gitea
      - POSTGRES_DB=gitea
    restart: always
    volumes:
      - gitea_postgres:/var/lib/postgresql/data
    expose:
      - 5432
    networks:
      - webnet

volumes:
  gitea_postgres:
    name: gitea_postgres
  gitea:
    name: gitea

networks:
  webnet:
    name: webnet
#     external: true


# Gitea is not allowed to webhook to Jenkins folow these steps
# 1) Go to Gitea Container
# 2) cat /data/gitea/conf/app.ini
```

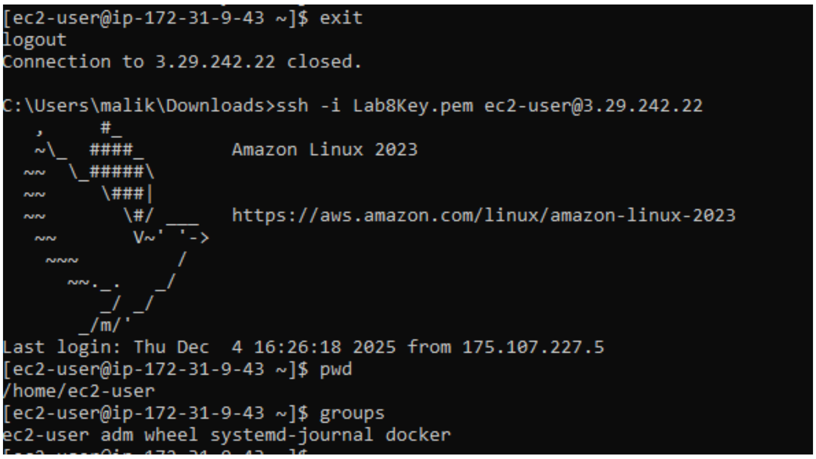
Step 8: Verify compose.yaml exists
Screenshot: task4_compose_yaml_saved_ls.png

```
[ec2-user@ip-172-31-9-43 ~]$ [ec2-user@ip-172-31-9-43 ~]$ ls -l
total 4
-rw-r--r--. 1 root root 1126 Dec  4 16:35 compose.yaml
[ec2-user@ip-172-31-9-43 ~]$ cat compose.yaml
services:
  gitea:
    image: gitea/gitea:latest
    container_name: gitea
    environment:
      - DB_TYPE=postgres
      - DB_HOST=db:5432
      - DB_NAME=gitea
      - DB_USER=gitea
      - DB_PASSWD=gitea
    restart: always
    volumes:
      - gitea:/data
    ports:
      - 3000:3000
    extra_hosts:
      - "www.jenkins.com:host-gateway"
    networks:
      - webnet
```

Step 9: Add ec2-user to docker group, show groups before and after reconnect
Screenshot: task4_usermod_and_groups_before_after.png

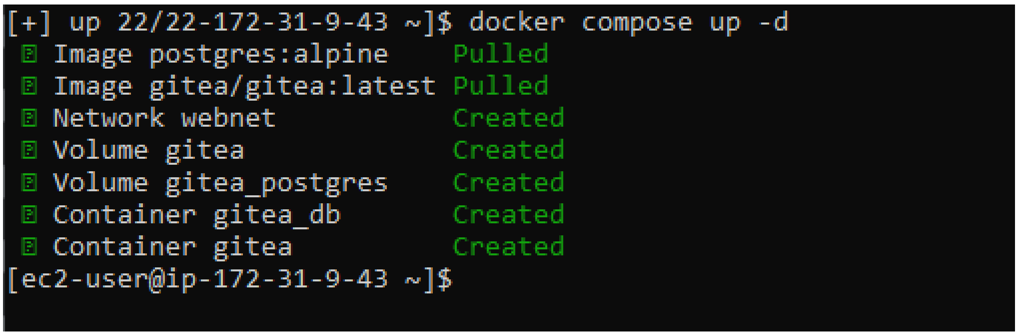


```
[ec2-user@ip-172-31-9-43 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-172-31-9-43 ~]$ sudo usermod -aG docker $USER
[ec2-user@ip-172-31-9-43 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-172-31-9-43 ~]$
```



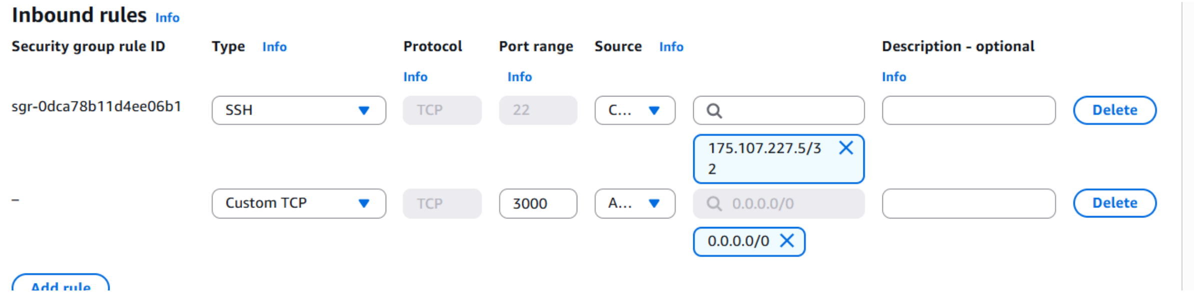
```
[ec2-user@ip-172-31-9-43 ~]$ exit
logout
Connection to 3.29.242.22 closed.

C:\Users\malik\Downloads>ssh -i Lab8Key.pem ec2-user@3.29.242.22
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Dec  4 16:26:18 2025 from 175.107.227.5
[ec2-user@ip-172-31-9-43 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-9-43 ~]$ groups
ec2-user adm wheel systemd-journal docker
```

Step 10: Run docker compose up -d
Screenshot: task4_docker_compose_up.png

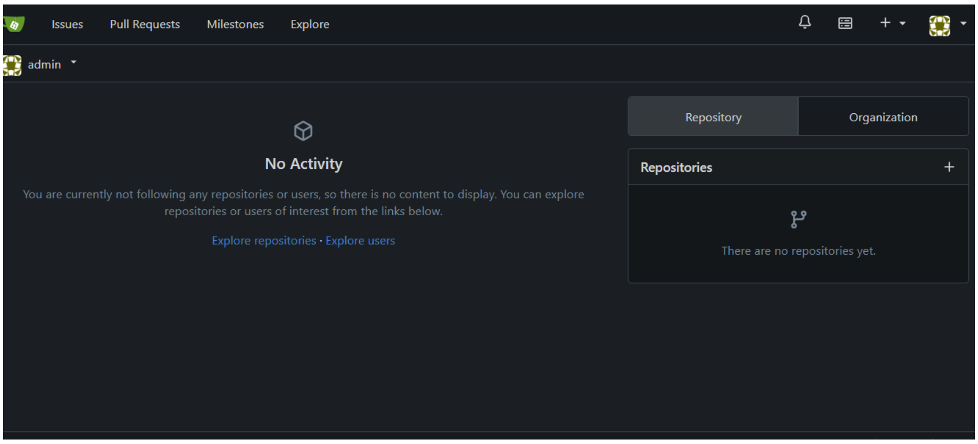


```
[+] up 22/22-172-31-9-43 ~]$ docker compose up -d
 Image postgres:alpine     Pulled
 Image gitea/gitea:latest  Pulled
 Network webnet            Created
 Volume gitea              Created
 Volume gitea_postgres     Created
 Container gitea_db        Created
 Container gitea           Created
[ec2-user@ip-172-31-9-43 ~]$
```

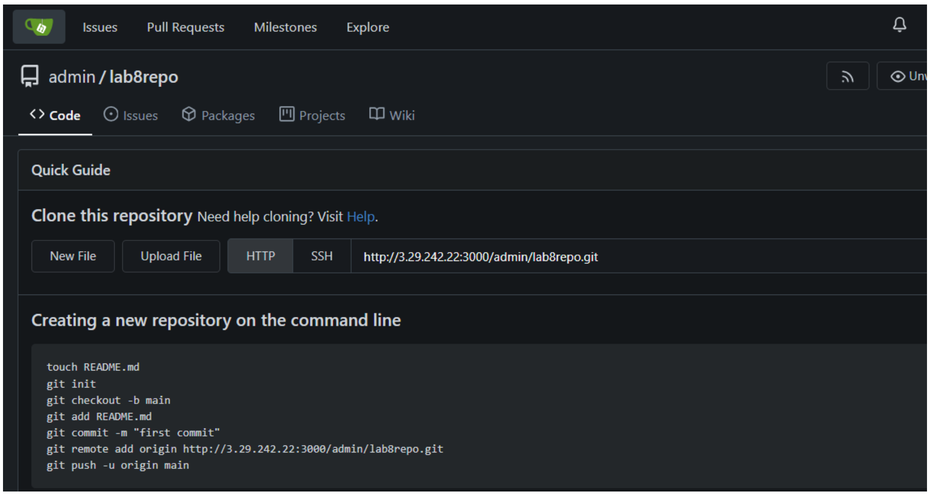
Step 11: Update Security Group Lab8SecurityGroup inbound rules to allow TCP 3000
Screenshot: task4_security_group_allow_3000.png



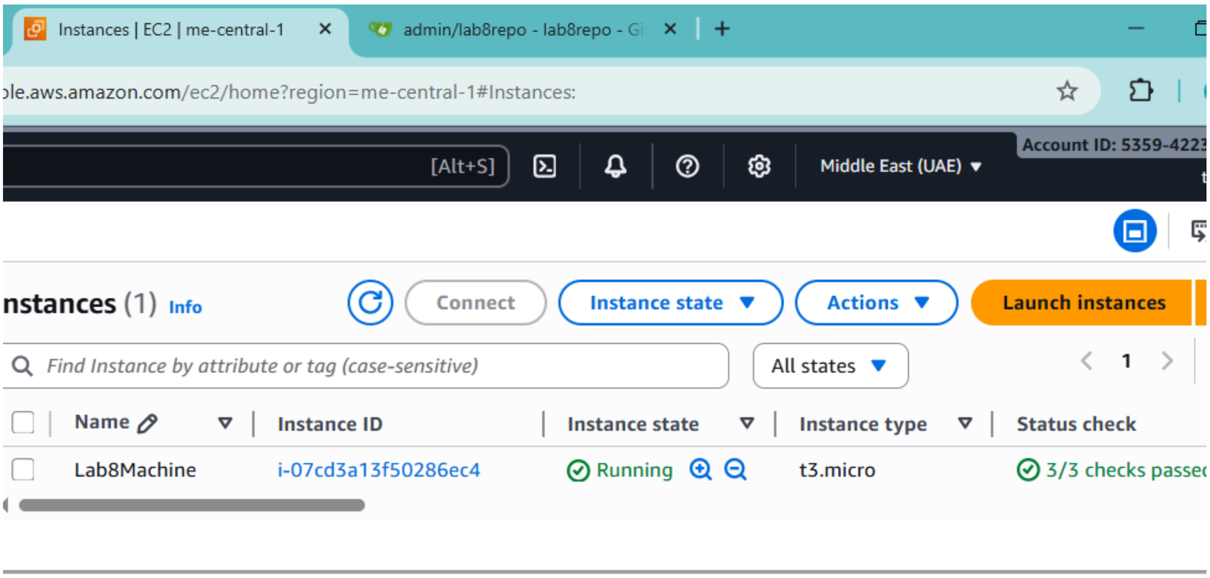
Step 12: Access Gitea installation page in browser
Screenshot: task4_gitea_install_page.png



Step 13: Complete Gitea setup and create repository
Screenshot: task4_gitea_create_repo.png



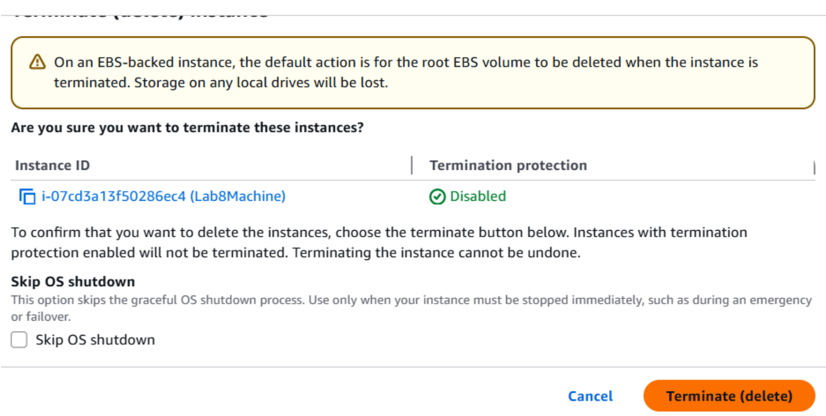
Step 14: Task 4 Summary — combined EC2, Security Group, Gitea UI
Screenshot: task4_summary.png



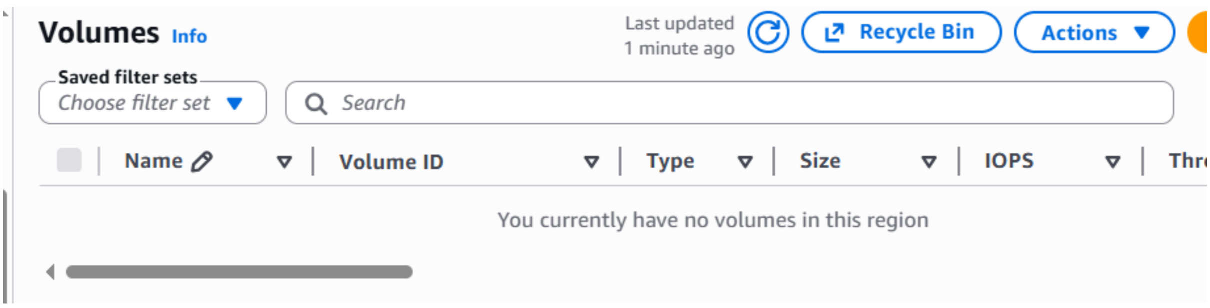
Cleanup — Remove All Resources

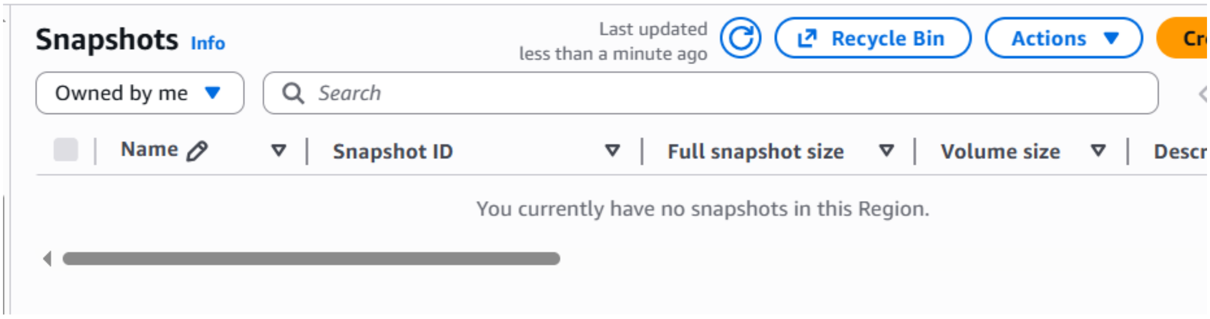
Goal: Terminate EC2, delete volumes, security groups, key pairs, IAM users.

Step 1: Terminate EC2 instance Lab8Machine
Screenshot: cleanup_terminate_instance.png

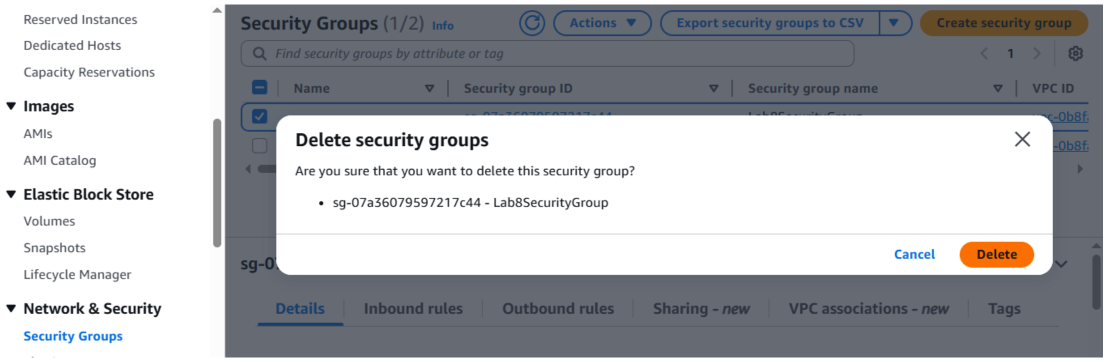


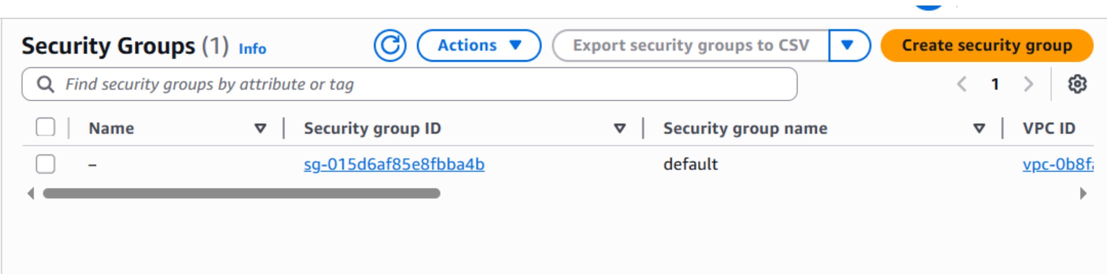
Step 2: Delete associated EBS volumes and snapshots
Screenshot: cleanup_delete_volumes_snapshots.png

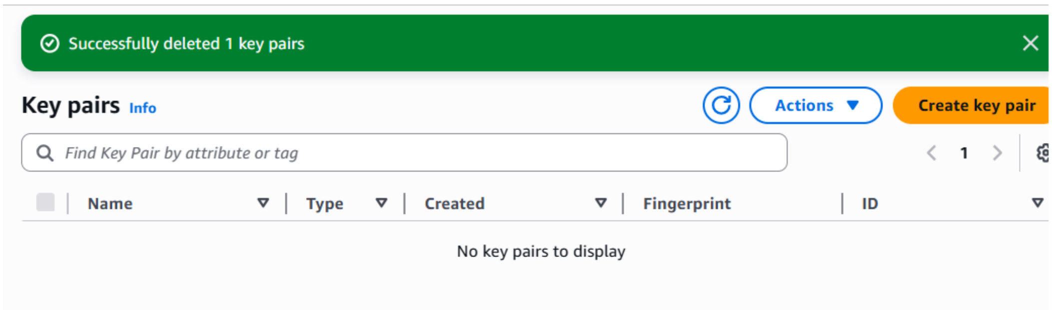




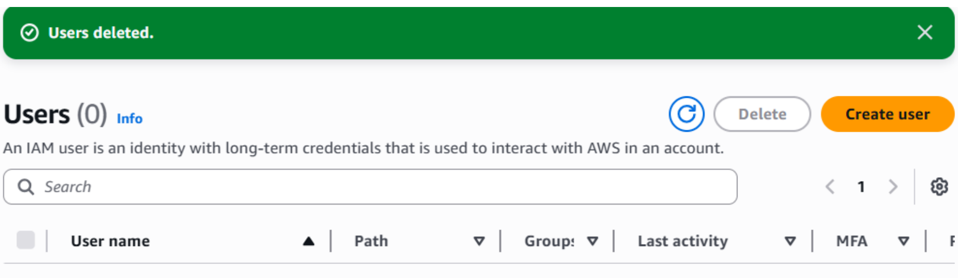
Step 3: Delete Security Group Lab8SecurityGroup and Key Pair Lab8Key
Screenshot: cleanup_delete_security_group_and_keypair.png







Step 4: Delete IAM users Lab8User and access keys
Screenshot: cleanup_iam_users_deleted.png



Step 5: Final cleanup summary — show billing/resources with no active resources
Screenshot: cleanup_summary.png